

IoT-Based Water Tank Monitoring System

Electronics & Communication Engineering Project Report

■ Project Overview

The IoT-Based Water Tank Monitoring System is an embedded electronics project designed to monitor the water level inside a tank using an ultrasonic sensor. The system uses an Arduino UNO to measure the distance between the sensor and the water surface and displays the water-level information on an I2C LCD.

■ Objectives

- Monitor water level automatically
- Reduce water overflow
- Display real-time water-level information
- Provide an affordable embedded solution
- Demonstrate sensor interfacing with a microcontroller

■ Components Used

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- 16×2 I2C LCD
- Buzzer/LED
- Jumper Wires
- Breadboard
- Power Supply

■ Software Used

- Arduino IDE
- Embedded C/C++
- Git
- GitHub

■■ Working Principle

The HC-SR04 ultrasonic sensor sends an ultrasonic pulse toward the water surface. The Arduino measures the time taken for the echo to return and calculates the distance. The measured water-level information is then displayed on the LCD.

■ System Block Diagram

Water Tank



HC-SR04 Sensor



Arduino UNO



LCD Display Buzzer/LED

■ Project Structure

IoT-Water-Tank-Monitoring/

■■■ Arduino_Code/

■■■ Circuit_Diagram/

■■■ Images/

■■■ Documentation/

■■■ README.md

■ Result

The system successfully detects the distance between the ultrasonic sensor and water surface and displays the measured value on the LCD.

■ Future Improvements

- ESP32 integration
- Wi-Fi connectivity
- Mobile application
- Cloud monitoring
- Automatic water pump control
- MQTT-based communication

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